

Autonomous Vehicle Testing



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CAVS Off-road Proving Ground



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NATURE Stack

Remote Access and Command and Control UI

Data Available to You

- Remote monitoring of vehicle behavior
 - Google Earth/Maps imagery
 - Vehicle position
 - Mission Waypoints
 - Global Path
 - Local Path
 - Occupancy Grid
 - Point Cloud
 - Camera
 - System Status
 - Are topics publishing? What rate?
 - Are major processing nodes operational?
- Remote command and control
 - Go/Stop/Shutdown buttons
 - Set mission/goal waypoints
- Real-time feedback while testing the system



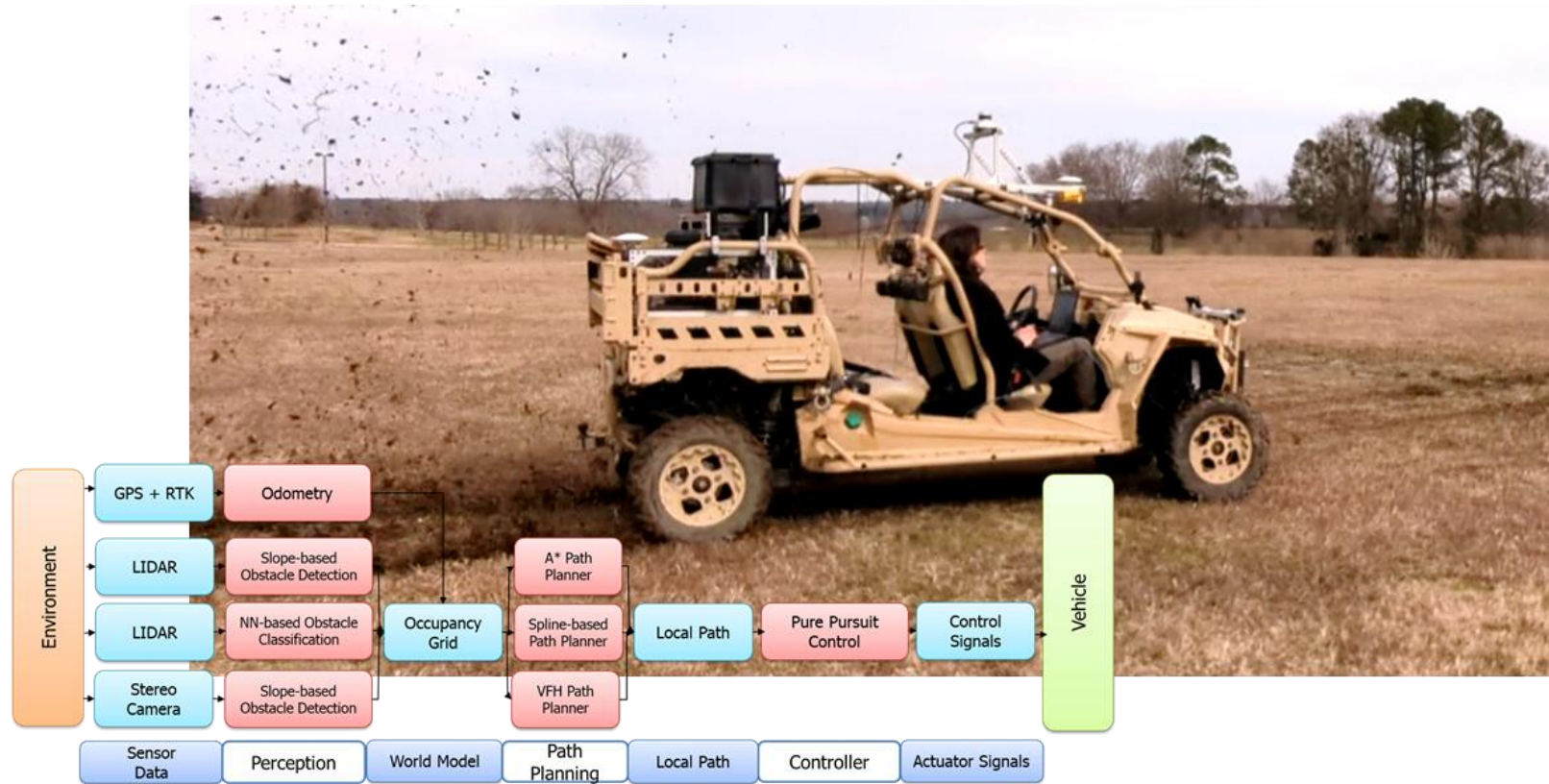
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NATURE Stack

Navigating All-Terrain Using Robotic Exploration



CAVS site: <https://www.cavs.msstate.edu/capabilities/nature.php>

Open source, available on Github: <https://github.com/CGoodin/nature-stack>



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Simulated Off-road Environment

Simulated off-road course behind your dataset

Physics-based sensor and vehicle simulation
Source of the ROS 2 bag you will work from
Faults are injected into that bag: GPS dropout,
LiDAR dropout, camera lag

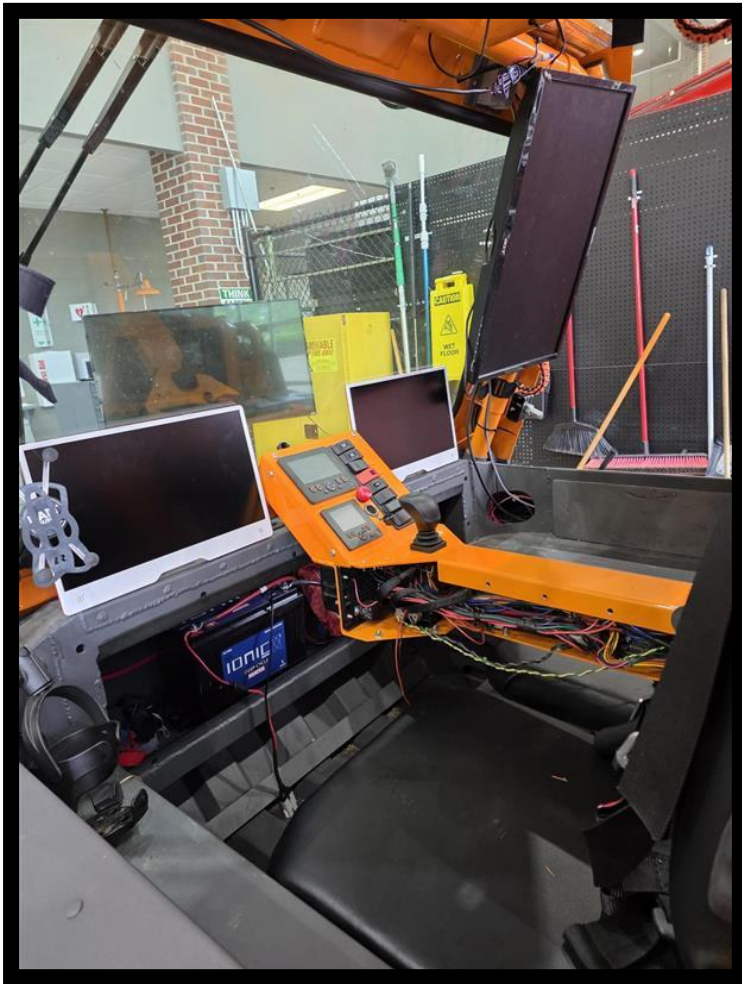


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Project - Overview



- Supervisory GUI for an off-road platform in ROS 2.
- Runs in-browser on a 12" tablet display driven by a Raspberry Pi 5.
- Provided ROS 2 bag now carries injected faults: GPS dropout, LiDAR dropout, camera lag.
- Each team decides early what problem it is solving — reviewing a completed run, a command-and-control handoff, or similar.
- Teams choose which data matters. We do not prescribe the display.



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Project 2 - Tools

- ROS 2 topics aligned with NATURE/MTX
- Robot Web Tools (ros2-web-bridge, roslibjs)
- ROS 2 bag playback (provided bag, faults injected)
- Tablet + Raspberry Pi 5 deploy toolchain



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